

Campylobacter Report: Results

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Exposure Computation

Table 1: Computation of exposure quantities

Behavior	Computation
Bathing	<code>bathing water vol · sample(EB)</code>
Drinking (breast milk)	<code>DrinkBM = breast milk vol · sample(HB)</code> <code>Areola = sample(HA)</code>
Drinking (water branch)	<code>drinking water vol · sample(ED)</code>
Mouthing-mom hand	<code>p_{hand→mouth} · sample(HP)</code>
Mouthing-other adult hand	<code>p_{hand→mouth} · sample(HP)</code>
Mouthing-child hand	<code>p_{hand→mouth} · sample(HH)</code>
Mouthing-fomite	<code>p_{fomite→mouth} · sample(EV)</code>
Eating (injera / other / raw plant)	<code>sample(EF) · 50</code>
Pica-soil	<code>sample(ES) · $\frac{1.25}{50}$</code>
Pica-other	0
Pica-feces	0

Notes:

- Drinking behavior consists of two branches: a breast milk branch (producing `num.bm` and `num.areola`) and a drinking water branch (producing `num.dw`), determined by a Bernoulli draw.
- All three eating behaviors (injera, other, raw plant) use the same formulation with a fixed multiplier of 50 representing serving size.
- The computation of `num.adult.hand` uses `mother handrinse` data rather than a separate adult-hand source.
- The behaviors `pica_other` and `pica_feces` do not have corresponding exposure terms in the implementation and are effectively assigned zero contribution.
- `phand→mouth` and `pfomite→mouth` are computed by functions $\propto \text{det.bact.hand.mouth}() * \text{fraction.hand.in.mouth}()$.

Questions for Campy Migration

1. Choice of dataset: `df_via` vs. `df_tot`

At present, two datasets are available, `df_via` and `df_tot`. It is not entirely clear what their biological or preprocessing differences are. In particular:

- What is the key distinction between these two datasets in terms of measurement?
- Which dataset is more appropriate for fitting the environmental concentration model in campy project?

2. High censoring rates and instability in posterior inference

When fitting the model using `df_via`, several sources (especially EF, EV, and HA) exhibit very high censoring rates, leading to unstable posterior distributions.

More importantly, for HB (breast milk), which is biologically expected to be an important pathway for *Campylobacter* transmission, there are no uncensored observations in `df_via`. After switching to `df_tot`, censoring rates decrease for most sources, but HB still remains highly censored (approximately 98%).

This raises several questions:

- Should HB still be included in the fitted model despite extreme censoring?
- Should it be modeled differently from other sources (e.g., with stronger priors or hierarchical structure)?
- Do we need to do sensitivity analysis to HB and other highly censored sources or handle them as insufficient direct data?

3. Use of a fixed multiplier (50 g) for all eating behaviors

In the exposure calculation, all eating-related behaviors multiply food concentration by a fixed value of 50, which appears to represent a 50 g serving size.

However, this assumption may not be realistic in all cases. In particular, when the simulated time interval is short, multiple eating events within a short period would imply repeated ingestion of 50 g each time, then is the computation appropriate?

4. Handling of `pica_other` and `pica_feces`

In the current implementation, no contamination values are assigned to `pica_other` and `pica_feces`, even though these behaviors are present in the dataset.

Is this because bacterial load data were not collected for these materials, or is there another modeling reason for assigning zero exposure to these pathways?

5. Units of concentration data

This is a quick question. I wonder whether the units of concentration are consistent with those used in the *E. coli* project. For example, is EF (food) still measured per gram?